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the ReceiptIQ project. My primary responsibilities included coordinating development across all modules, ensuring seamless integration between components, managing project timelines, and supervising the final deployment of our privacy-first personal finance assistant.

System Architecture & Design

The Modular Pipeline

A core part of my role was designing and overseeing the high-level architecture of the system to ensure it was scalable, maintainable, and easy to improve. I structured ReceiptIQ into a clear, modular pipeline:

1. **OCR Engine:** Extracts raw text from receipt images.
2. **Preprocessing Layer:** Cleans and structures the extracted text.
3. **ML Classifier:** Predicts the expense category using trained models.
4. **Backend Layer:** Manages transactions and local storage.
5. **Analytics Engine:** Produces spending patterns and forecasts.
6. **Frontend Interface:** Provides a clean user interaction layer via Gradio.

By enforcing this modularity, each team member could develop and test their components independently before final integration.

Key Technical Decisions

Throughout the project lifecycle, I guided the team in making strategic technical choices that aligned with our core philosophy: **Privacy First**. These key decisions included:

- **EasyOCR over Cloud APIs:** Ensuring no sensitive financial documents are ever sent to third-party servers.
- **TF-IDF + Classical ML:** Opting for lightweight local inference that runs quickly on consumer hardware without requiring heavy compute resources.
- **Gradio UI:** Allowing for rapid deployment and a clean, accessible interaction model for non-technical users.
- **Local-First Storage:** Guaranteeing full user ownership of their financial data with zero forced account creation or cloud dependencies.

Project Coordination & Integration

Beyond technical design, my contribution involved active project management. I facilitated communication between the Data Engineering (OCR) and Machine Learning teams to ensure the structured data output matched the required input features for the classification models. Furthermore, I worked closely with the

2. **Privacy as a feature:** Building a strictly local AI application introduces constraints (e.g., lightweight models, local storage), but ultimately delivers a highly trustworthy product.
3. **Cross-functional communication:** The success of an ML pipeline depends heavily on the data handoffs between different specialists; clear API contracts and data structures are essential.
4. **User-centric design:** Technical sophistication (like TF-IDF or Isolation Forests) must be hidden behind an intuitive, fast, and simple user interface to drive adoption.

Conclusion

I led the architectural design and overall execution of the ReceiptIQ pipeline. By establishing a robust modular framework and enforcing a strict privacy-first technical strategy, I successfully coordinated our team to transform raw receipt images into actionable financial intelligence. The resulting application automates expense tracking securely and efficiently.

Individual Contribution Statement: This report describes my independent contribution (Team Leadership, Architecture Design, and Project Management) to the group project.

Architecture & Planning

- **Core Blueprint:** Architected the end-to-end modular pipeline connecting EasyOCR extraction to ML categorization and Analytics.
- **Privacy First:** Formulated the decision to use 100% local inference, ensuring zero cloud-dependency for sensitive financial data.
- **Scoping:** Defined the technical roadmap and task distribution for the 5-member development team.

```
// Integration Logic
class ReceiptIQ_Hub:
    def initialize_pipeline(self):
        self.ocr = DataExtractor()
        self.ml = ExpenseClassifier()
        self.analytics = InsightEngine()
        self.ui = GradioInterface()
```

Repository & Flow Control

- **GitHub Management:** Acted as the gatekeeper for the repository, managing merge requests and ensuring version consistency.
- **State Handling:** Engineered the central backend logic to maintain user session data and prevent duplicate entries.
- **Agile Leadership:** Conducted technical stand-ups to resolve "hustles" and blockers within individual modules.

```
# Repo Structure Gatekeeping  
$ git checkout -b feature/integration  
$ git merge data-extraction  
$ git merge ml-categorization  
$ git merge user-analytics  
# All modules synchronized
```

Final Integration & Testing

- **System Synthesis:** Merged distinct technical modules into a single unified application interface.
- **Stress Testing:** Managed the QA process to verify OCR accuracy across various receipt qualities and formats.
- **Documentation:** Finalized technical documentation for project submission and future maintainability.

"The role of the Lead is to turn five separate voices into one clear, powerful project. ReceiptIQ is the result of that synergy."